

DE QUEEN SEVIER COUNTY AP, AR, USA

WMO: 743312

Lat: 34.047N

Lon: 94.399W

Elev: 108

StdP: 100.03

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-4.6	-12.8	1.3	-1.2	-10.3	1.6	0.8	8.4	11.9	7.5	10.2	0.9	340	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	37.1	24.6	35.6	24.7	34.1	24.7	26.9	32.2	26.4	32.0	25.8	31.4	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.2	28.9	25.0	20.3	28.6	24.4	19.6	28.2	84.9	32.1	82.4	31.8	80.3	31.4	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.8	5.1	-10.3	39.0	2.2	2.3	-11.9	40.7	-13.1	42.1	-14.4	43.4	-15.9	45.1		
			-10.8	27.9	2.1	0.8	-12.2	28.5	-13.5	28.9	-14.6	29.4	-16.1	30.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.9	7.7	12.7	16.7	21.3	25.9	27.4	27.7	24.1	17.6	11.6	7.2	(6)
		DBStd	8.76	5.11	5.14	4.88	4.01	3.51	2.07	2.08	2.28	3.15	4.61	4.95	4.85	(7)
		HDD10.0	436	148	98	29	3	0	0	0	0	3	38	117		(8)
		HDD18.3	1587	386	299	183	77	15	0	0	2	71	207	347		(9)
		CDD10.0	3057	21	33	112	203	350	476	539	548	422	238	87	29	(10)
		CDD18.3	1168	1	1	8	26	106	226	281	289	173	48	6	1	(11)
		CDH23.3	12298	2	8	81	281	968	2347	3107	3212	1753	486	50	4	(12)
		CDH26.7	5776	0	1	6	42	298	1083	1628	1732	829	151	7	0	(13)
(14)	Wind	WSAvg	1.8	2.0	2.2	2.4	2.2	1.8	1.6	1.5	1.4	1.4	1.5	1.7	1.9	(14)
(15)	Precipitation	PrecAvg	1396	117	103	120	132	153	111	97	71	112	128	117	129	(15)
(16)		PrecMax	1883	200	229	269	295	345	233	218	207	261	328	347	228	(16)
(17)		PrecMin	819	14	34	19	52	65	41	6	10	1	0	37	9	(17)
(18)		PrecStd	277	52	47	63	60	70	57	62	44	61	91	79	59	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	23.0	24.6	27.5	29.8	32.8	36.1	38.6	40.2	37.1	32.3	27.5	23.7		(19)
(20)		MCWB	15.8	17.3	18.5	20.5	23.1	24.3	24.6	24.3	23.4	22.4	20.3	18.5		(20)
(21)		DB	20.8	22.2	25.6	27.8	31.1	34.3	36.7	37.4	34.8	30.3	24.6	21.2		(21)
(22)		MCWB	15.7	16.1	17.7	19.3	23.0	24.7	24.7	24.7	23.2	21.9	17.9	16.9		(22)
(23)		DB	18.7	20.0	23.8	26.4	29.8	33.1	35.1	36.0	33.1	28.3	22.4	19.0		(23)
(24)		MCWB	15.1	15.0	17.1	19.3	22.7	24.7	25.2	24.9	23.5	20.4	17.7	15.1		(24)
(25)		DB	15.7	17.6	21.9	24.9	28.3	32.0	33.7	34.3	31.5	26.2	20.5	16.5		(25)
(26)		MCWB	11.0	13.5	16.0	18.4	21.9	24.5	25.2	25.0	22.8	19.7	16.4	13.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	18.7	20.1	20.6	22.9	25.4	27.2	27.5	27.8	26.1	24.6	21.9	20.1		(27)
(28)		MCDB	20.7	22.3	24.7	26.8	30.0	32.0	32.5	32.7	31.4	29.4	25.0	22.2		(28)
(29)		WB	17.6	18.1	19.5	21.6	24.4	26.3	26.8	26.9	25.4	23.7	20.4	18.1		(29)
(30)		MCDB	19.8	20.4	23.6	25.7	29.0	31.6	32.4	32.3	31.2	28.0	22.9	20.5		(30)
(31)		WB	15.3	16.4	18.4	20.7	23.7	25.7	26.3	26.4	24.8	22.7	18.8	15.8		(31)
(32)		MCDB	17.7	19.1	22.5	24.6	28.1	30.9	32.2	32.0	30.2	26.3	21.6	18.1		(32)
(33)		WB	12.1	14.0	17.1	19.8	22.9	25.2	25.8	25.8	24.0	21.2	17.2	13.5		(33)
(34)		MCDB	14.4	16.4	20.8	23.4	27.0	30.3	31.4	31.5	28.9	24.2	19.8	15.6		(34)
(35)	Mean Daily Temperature Range	MDBR	12.7	12.2	13.3	13.2	11.7	11.4	11.8	12.4	13.3	13.9	13.5	12.4		(35)
(36)		5% DB	16.3	16.4	14.9	14.7	13.1	13.0	14.2	15.2	15.1	15.7	15.2	15.6		(36)
(37)		MCWBR	10.9	10.9	8.4	7.7	6.0	5.0	4.2	4.3	5.2	7.5	9.4	10.8		(37)
(38)		5% WB	12.3	12.9	12.2	11.4	10.9	11.2	11.7	12.2	12.3	11.9	12.1	12.3		(38)
(39)		MCWBR	9.9	9.8	7.6	7.1	5.7	4.8	4.6	4.6	4.9	6.5	9.0	9.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.324	0.330	0.361	0.396	0.430	0.451	0.469	0.468	0.405	0.360	0.337	0.319		(40)
(41)		taud	2.488	2.476	2.413	2.315	2.283	2.263	2.242	2.232	2.372	2.479	2.494	2.514		(41)
(42)		Ebn at Noon	887	919	913	891	860	839	820	816	857	875	863	871		(42)
(43)		Edh at Noon	86	96	111	128	134	136	139	138	115	95	85	79		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.60	2.97	4.15	5.13	5.63	6.29	6.25	5.82	4.93	3.85	2.76	2.25		(44)
(45)		RadStd	0.24	0.45	0.32	0.32	0.52	0.63	0.41	0.45	0.43	0.54	0.34	0.29		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.07	+1.53	N/A	N/A	N/A	N/A	(46)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page